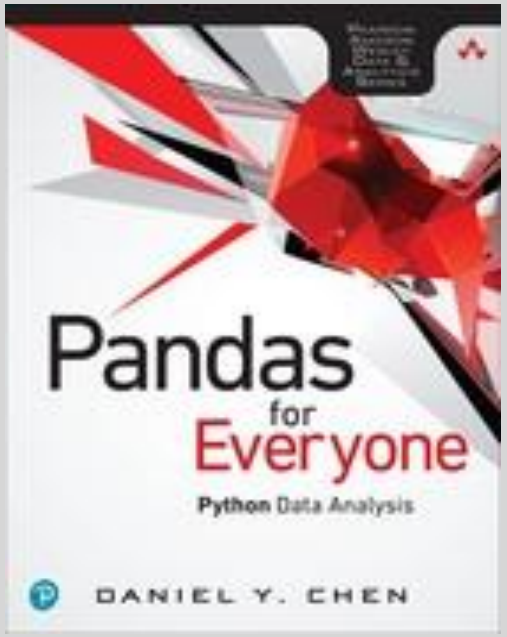
References


Data Engineering
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Martin Kleppmann


Sinan Ozdemir, Divya Susarla
Feature
Engineering
Made Easy
Packt


Pandas
for
Everyone
Python Data Analysis
DANIEL Y. CHEN

What is Data Engineering

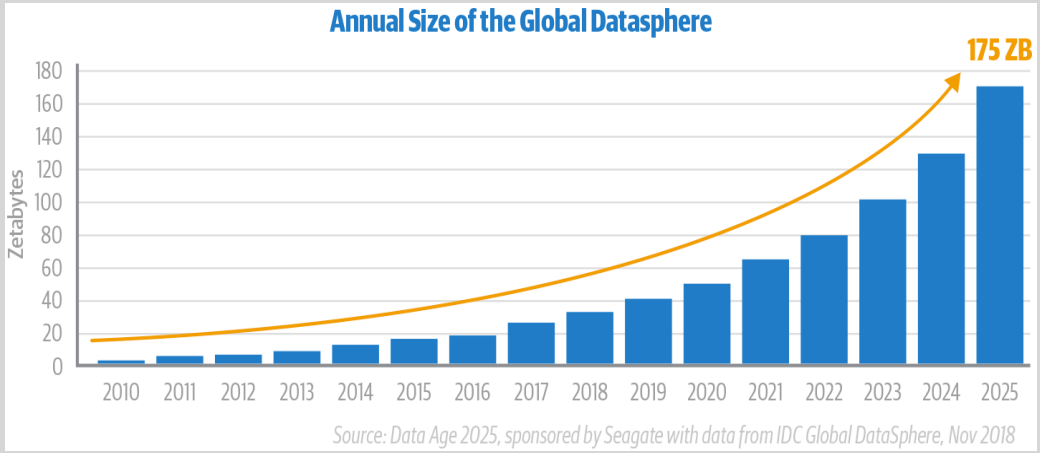
Data engineering is the process of designing, building, and maintaining the infrastructure and systems that collect, store, and process data.



Data Engineering Today

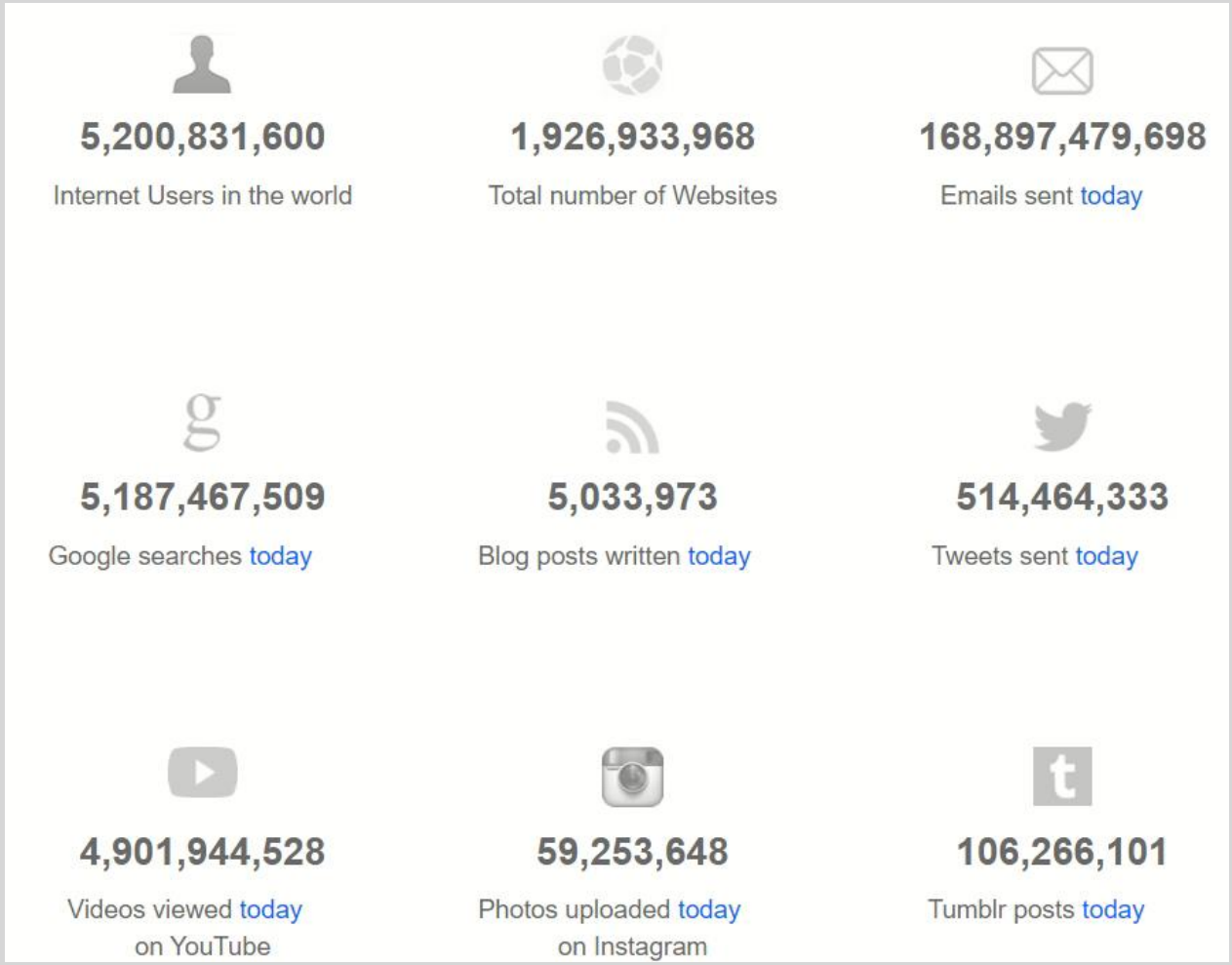
- Data Engineering → a critical component of data science and analytics.
  - provides the foundation for data-driven decision-making.

- With the GROWING amount of data businesses generates, data engineering has become an increasingly important field.



Unit	Abbr	Size (Decimal)
Kilobyte	Kb	10^3
Megabyte	Mb	10^6
Gigabyte	Gb	10^9
Terabyte	Tb	10^{12}
Petabyte	Pb	10^{15}
Exabyte	Eb	10^{18}
Zettabyte	Zb	10^{21}
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Data Engineering Today



Source

Data Engineering Today

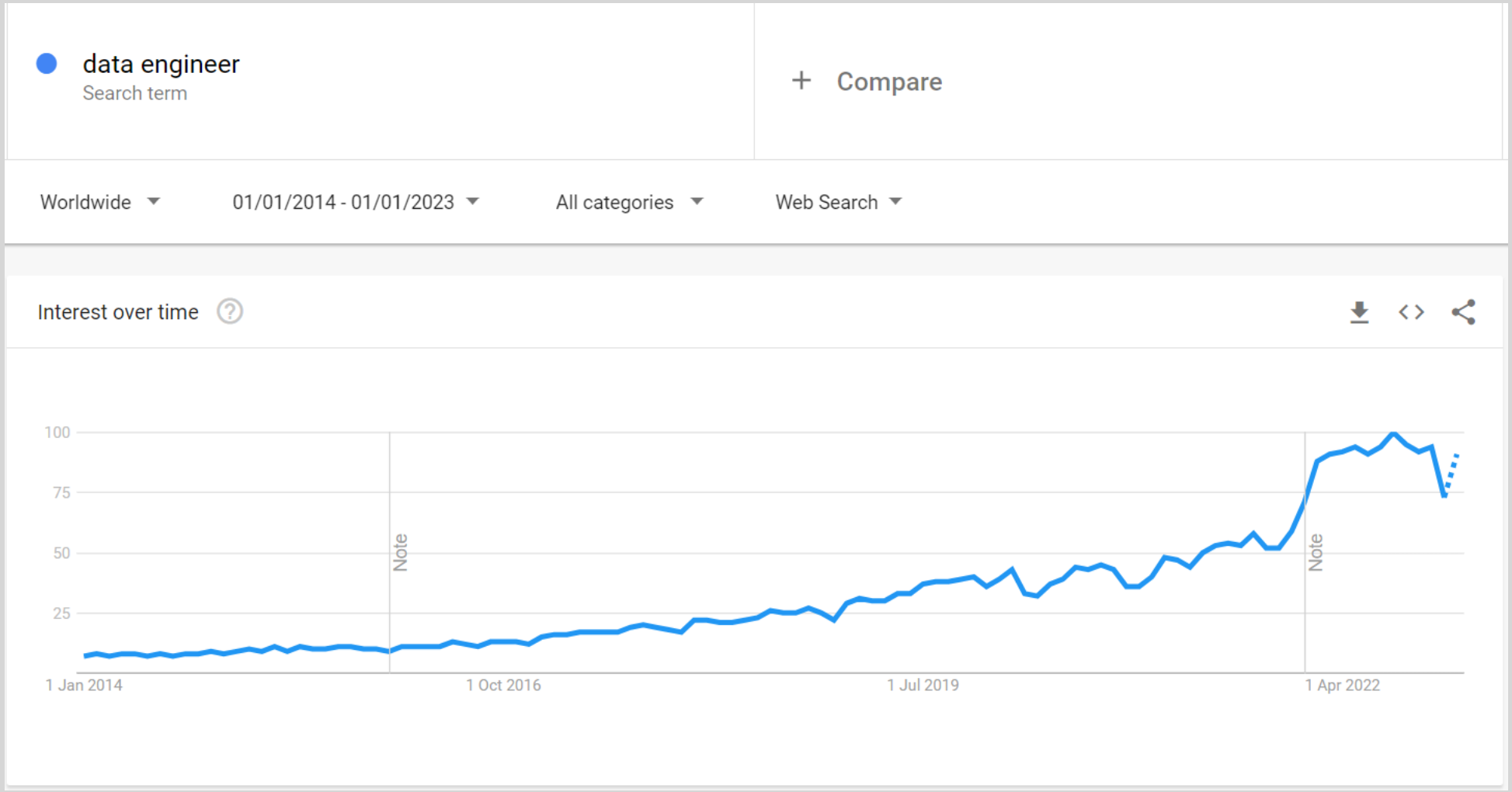
Data engineers→ database administrators (DBAs), data analysts, business intelligence engineers, and database developers.

The growth in data production has led to major advancements in data technology and data jobs, especially in the fields of data engineering and data science.



Today → Data engineering is about the movement, manipulation, and management of data.

Data Engineering Trend



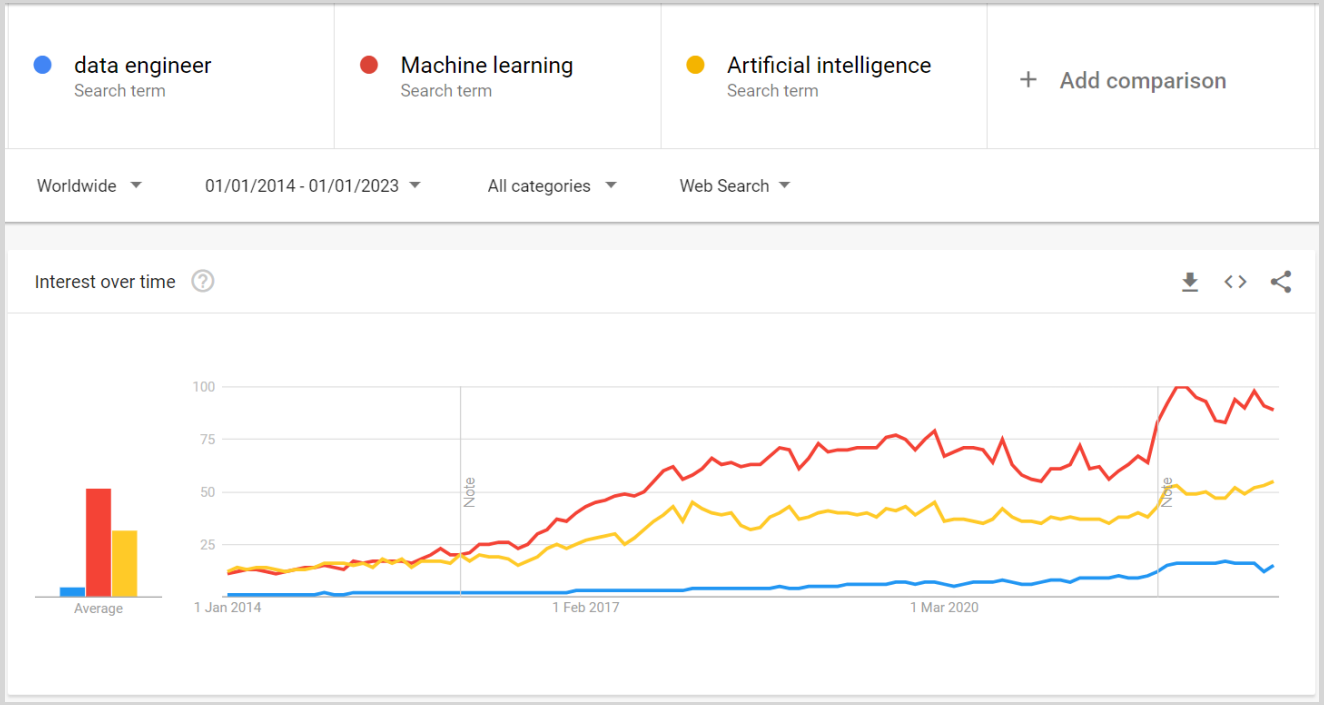
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Data Engineering Trend

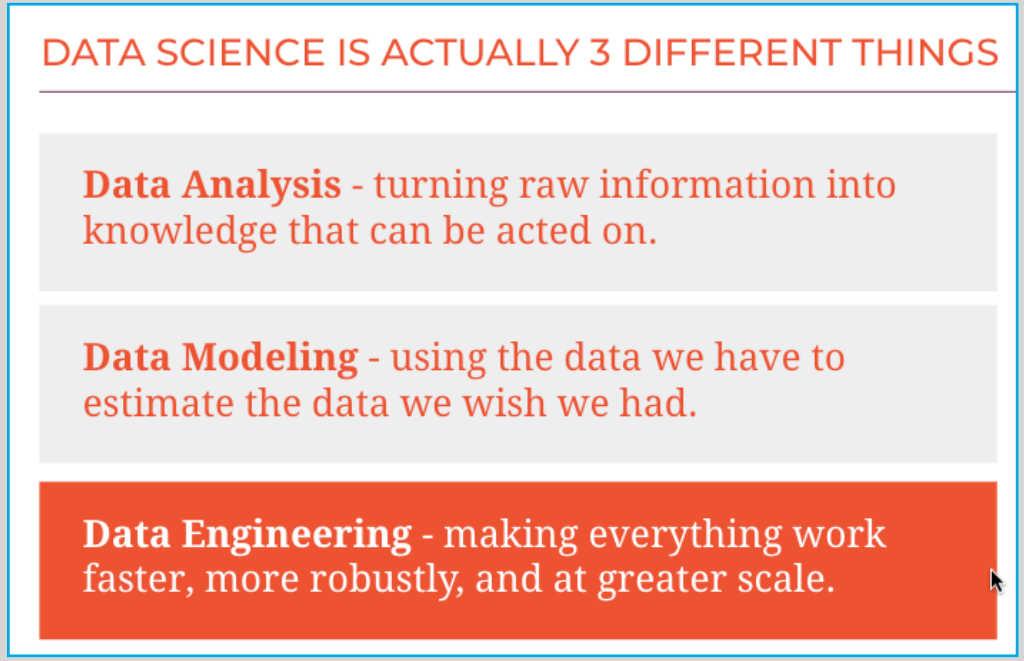
- Topics like machine learning and AI will always win a popularity contest, especially as they continue to appear in the mainstream media.



A good chunk of the work behind these concepts stems from data engineering work, such as data mining, data manipulation and cleaning, string manipulation, aggregation, joining to other data sets, and building deployable machine learning pipelines.



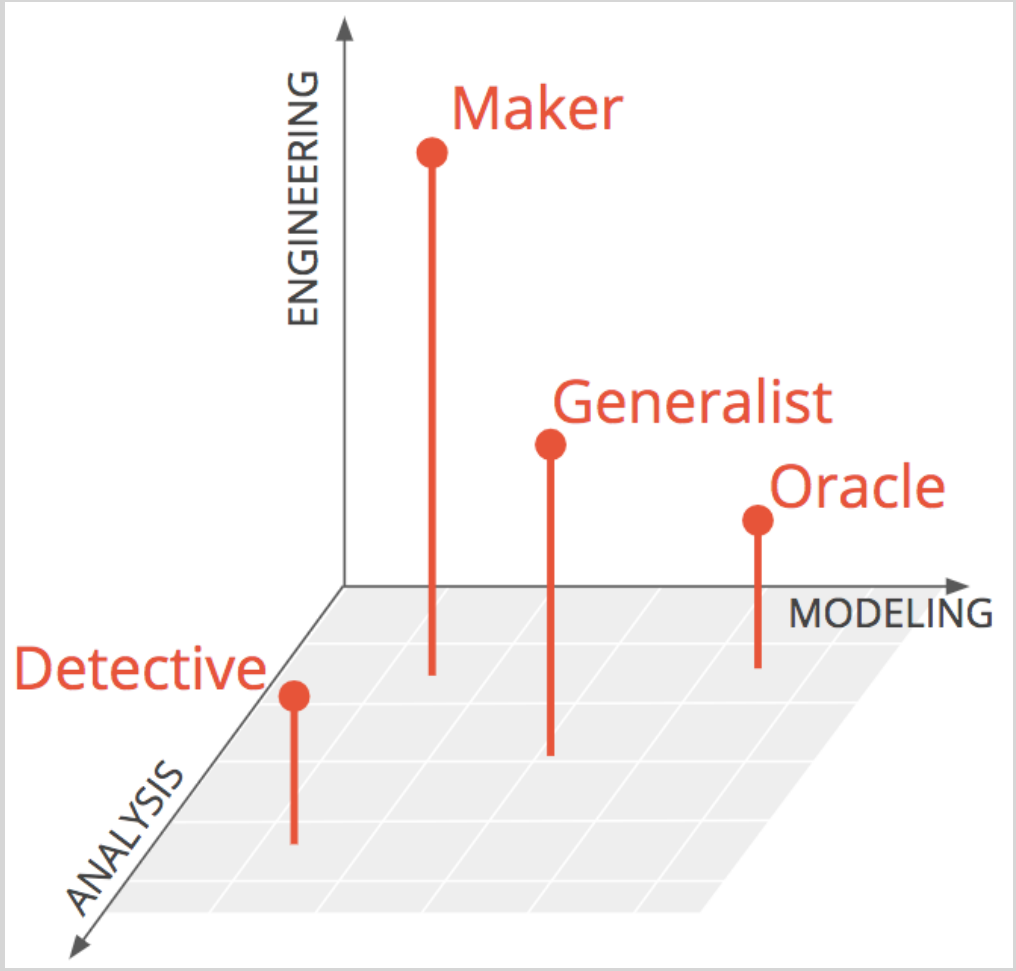
Data Science vs Data Engineering



- Data Science is the field of study that combines domain expertise, programming skills, and knowledge of mathematics and statistics to extract meaningful insights from data.

- Data engineering deals with **various data formats**, **storage**, **data extraction**, and **transformation**.

Data Science vs Data Engineering



DATA SCIENTIST ARCHETYPES

- **Generalist** (proficient at everything)
- **Detective** (mastery of analysis)
- **Oracle** (mastery of modeling)
- **Maker** (mastery of engineering)

Source

Data Scientist vs Data Engineer

What are the differences between Data scientists and Data engineers?

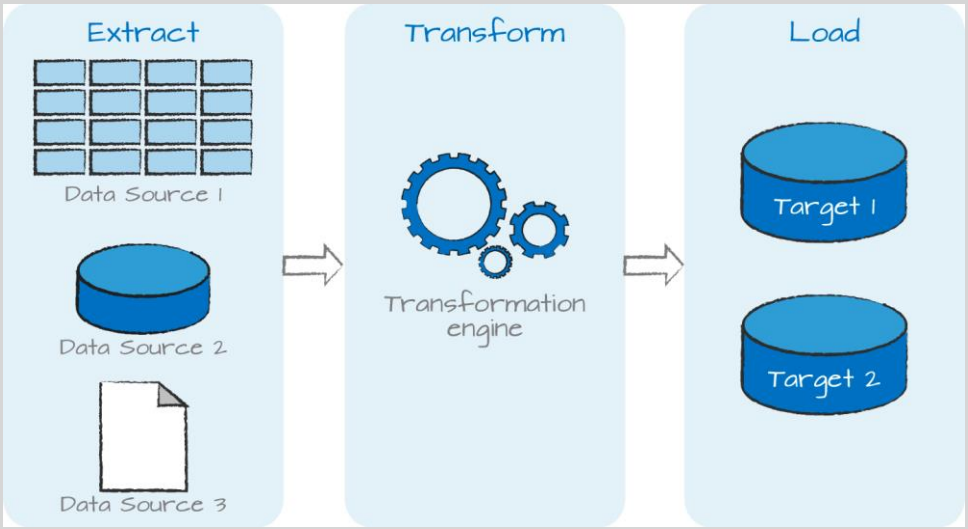
Data Scientist	Data Engineer
• More mathematical. • Utilize data for building statistical models and mathematical computation. • Their models incorporated into a data engineering pipeline. • Connect to the data warehouses built by data engineers.	• More technical with solid data warehousing and programming backgrounds. • Need to understand data formats, models, and structures to transport data efficiently. • Build data engineering pipeline.

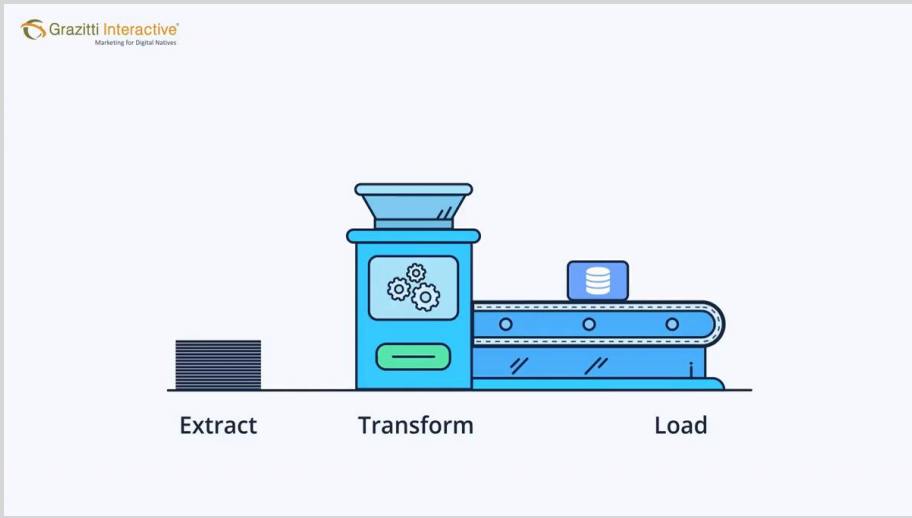
What Data Engineers Do





- ETL (extract, transform, and load) systems.
  - Data engineering involves the movement of data from one system or format to another system or format.
  - Data engineers query data from a source (EXTRACT), perform some modifications to the data (TRANSFORM) and then put that data in a location where users can access it and know that it is production quality (LOAD).



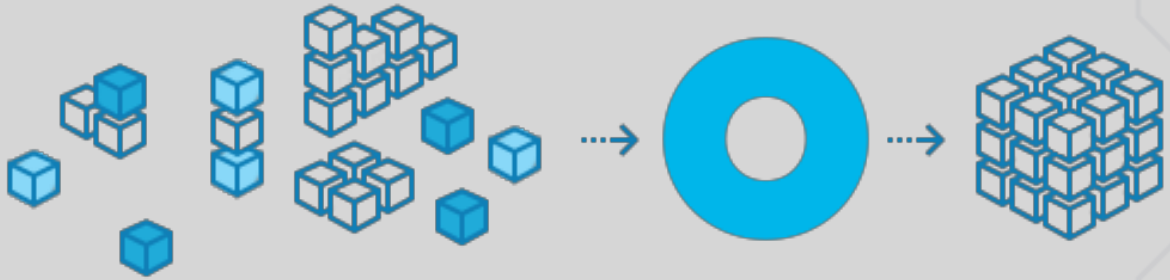




Data Extraction

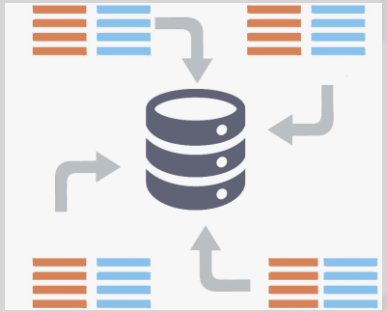
- Extract Data → Getting Data
- Data is copied or exported from source locations to a staging area.
- The data can come from any structured or unstructured source—SQL or NoSQL servers, CRM and ERP systems, text and document files, emails, web pages, and more.
- Data engineers dedicate a lot of their time to pulling data sets from an array of sources, most of the time into a central hub, a data warehouse, to be utilized together.

Data Transformation



- 80% of data science work is data preparation; 75% of data scientists find this the most boring aspect of the job.
- Raw data is transformed to be useful for analysis and to fit the schema of the eventual target data warehouse.
- Data engineer brings their skill in manipulating data to a project, such as:
  - Filtering, cleansing, de-duplicating, validating and authenticating the data.
  - Performing calculations, translations, or summaries based on the raw data.
  - Formatting the data into tables or joined tables to match the schema of the target data warehouse.

Data Loading



- Load is one of the more basic data engineering job functions because it is just the movement and storage of data (Store data to the target location).
- Data Engineers sometimes swap the load and transform steps around to be (ELT) when dealing with big data technologies such as Hadoop/Spark because the extraction process is cheaper to run and spreads the processing burden across multiple machines (cluster).
- Data engineers have the experience to choose from many available data storage options in the cloud and on-premise, including NoSQL databases, relational data stores, and data warehouses (Data lake).

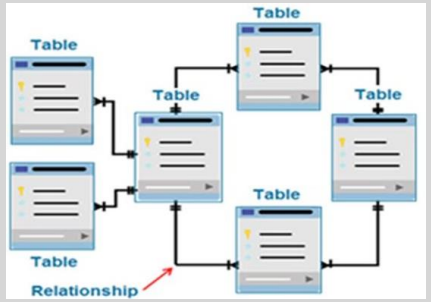
What Data Engineers Do





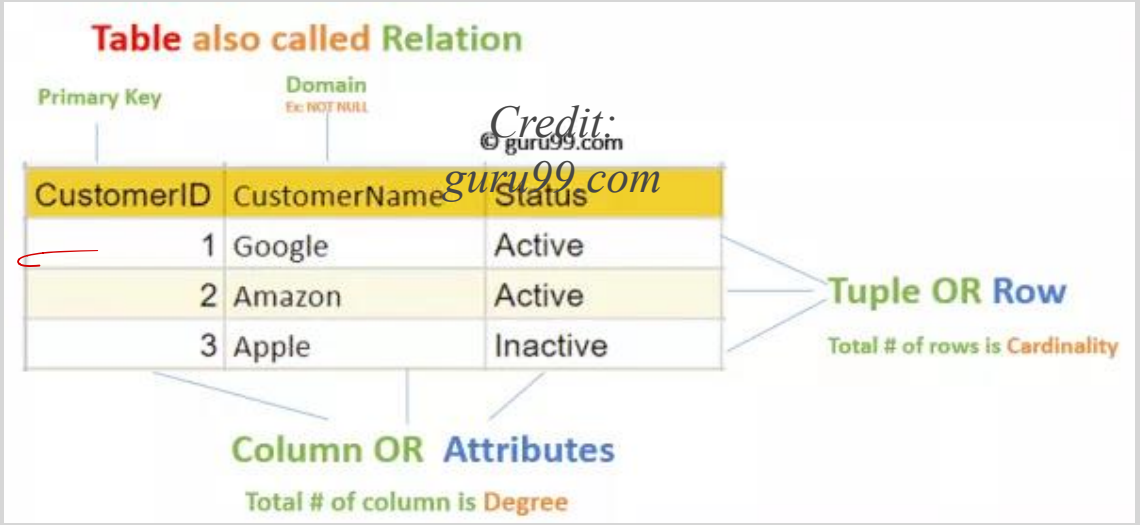
- Data storage.
  - Data engineers involve in data storage selection from several available cloud-based and on-premise options.
  - Relational and non-relational databases. Databases rank among the most common solutions for data storage. Data engineers should be familiar with relational and non-relational databases and how they work.
  - Data lake is a centralized repository that allows for storing structured and unstructured data at any scale.
  - Cloud computing → separates storage and computational machines, meaning you can scale down (switch off) expensive machines used to process data without affecting the stored data.

Relational Databases

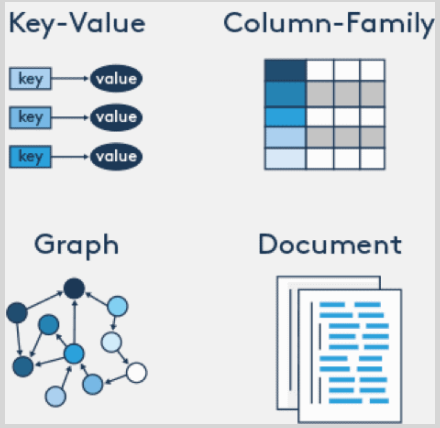


Relational database is a collection of data items with pre-defined relationships between them.

- These items are organized as a set of tables with columns and rows.
- Tables are used to hold information about objects.



Non-relational Databases



Non-relational database is a database that does not use the tabular schema of rows and columns. NoSQL stands for not only SQL. Google's Bigtable is an example of a NoSQL data store built on the Google File System (GFS).

- Documents
- Semi-structured data
- Large and unstructured data → Results from the internet of things (IoT), social networks, and the rise of AI.

- NoSQL databases have been popular because distributed technologies and file systems (including Hadoop/Spark) became more accessible for storing petabytes of data.
  - Distributed technology is a network of machines called a cluster, and each machine is referred to as a node.

Relational vs Non-relational Databases

Relational Database	Non-relational Database
• Use a structured schema to organize data into tables with rows and columns. • Use SQL (Structured Query Language) to access and manipulate the data. • Relationships between tables are established using foreign keys. • Ex. MySQL and PostgreSQL.	• Use a more flexible schema-less structure to store data in documents or key-value pairs. • Use their query languages. • Do not rely on foreign keys to establish relationships between data. • Used for storing large amounts of unstructured data. • Ex., MongoDB and Cassandra.

Data Warehouse

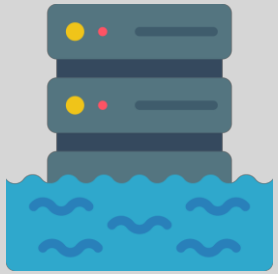


- Data warehouse is a large, centralized repository of data designed to support business intelligence activities, such as reporting and data analysis.

- Data is extracted from various transactional systems, transformed to fit a common data model, and then loaded into the warehouse for reporting and analysis.

- Data warehousing technologies can be implemented with relational databases, columnar databases or cloud-based services like Amazon Redshift, Snowflake and Google BigQuery.

Data Lake



- Data lake is a centralized, shared repository that allows storing all structured and unstructured data at any scale.

- Designed to handle large amounts of data, including raw, detailed, and historical data, and to support a wide variety of data processing and analysis tasks, such as batch processing, real-time streaming, interactive querying, dashboards, and visualizations of data.
  - Store data in its native format, WITHOUT having to fit a data model.

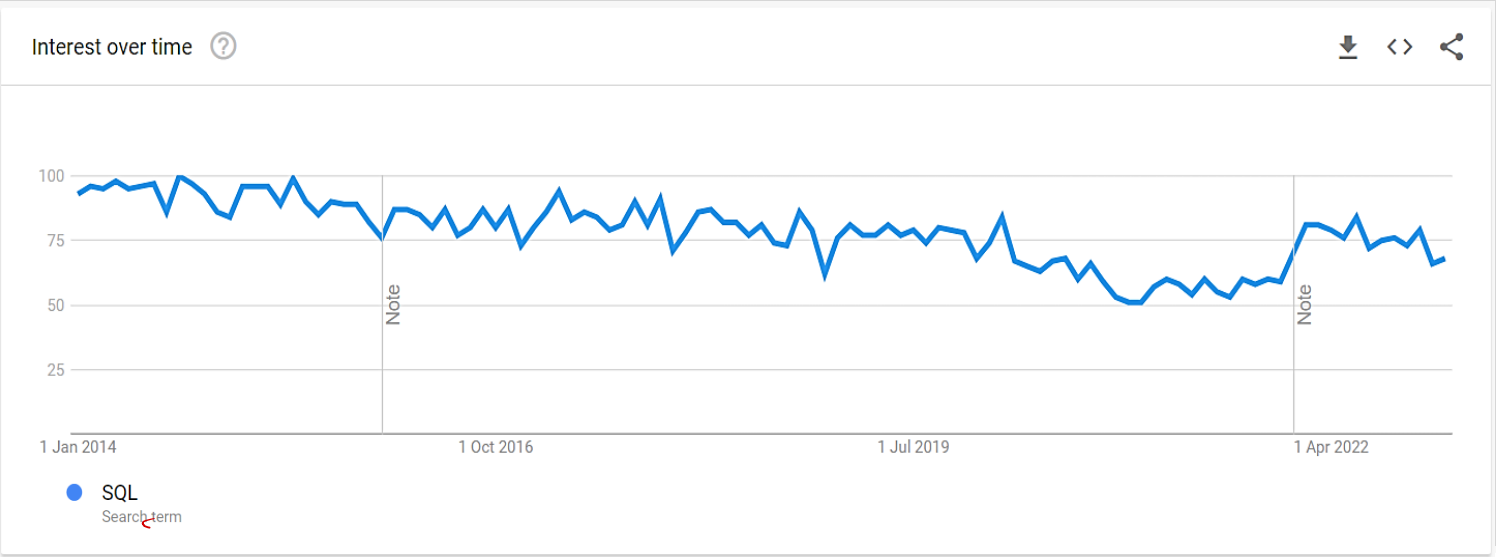
- Data lake technologies can be implemented on-premises or on cloud-based services like AWS S3, Azure Data Lake Storage, and Google Cloud Storage.

Data Warehouse vs Data Lake

Characteristics	Data Warehouse	Data Lake
Data	Relational from transactional systems, operational databases, and line of business applications	Non-relational and relational from IoT devices, web sites, mobile apps, social media, and corporate applications
Schema	Designed prior to the implementation (schema-on-write)	Written at the time of analysis (schema-on-read)
Price/Performance	Fastest query results using higher cost storage	Query results getting faster using low-cost storage
Data Quality	Highly curated data that serves as the central version of the truth	Any data that may or may not be curated (i.e., raw data)
Users	Business analysts	Data scientists, Data developers, and Business analysts (using curated data)
Analytics	Batch reporting, BI and visualizations	Machine Learning, Predictive analytics, data discovery and profiling

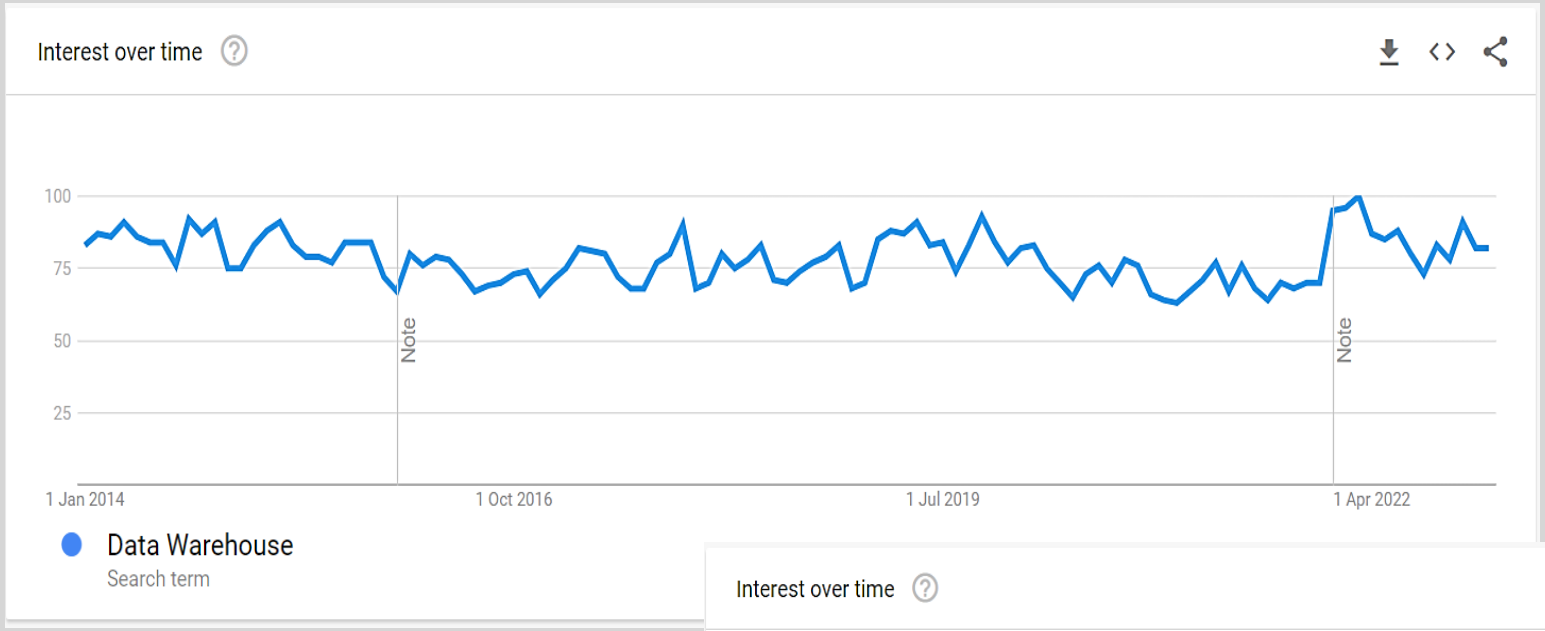
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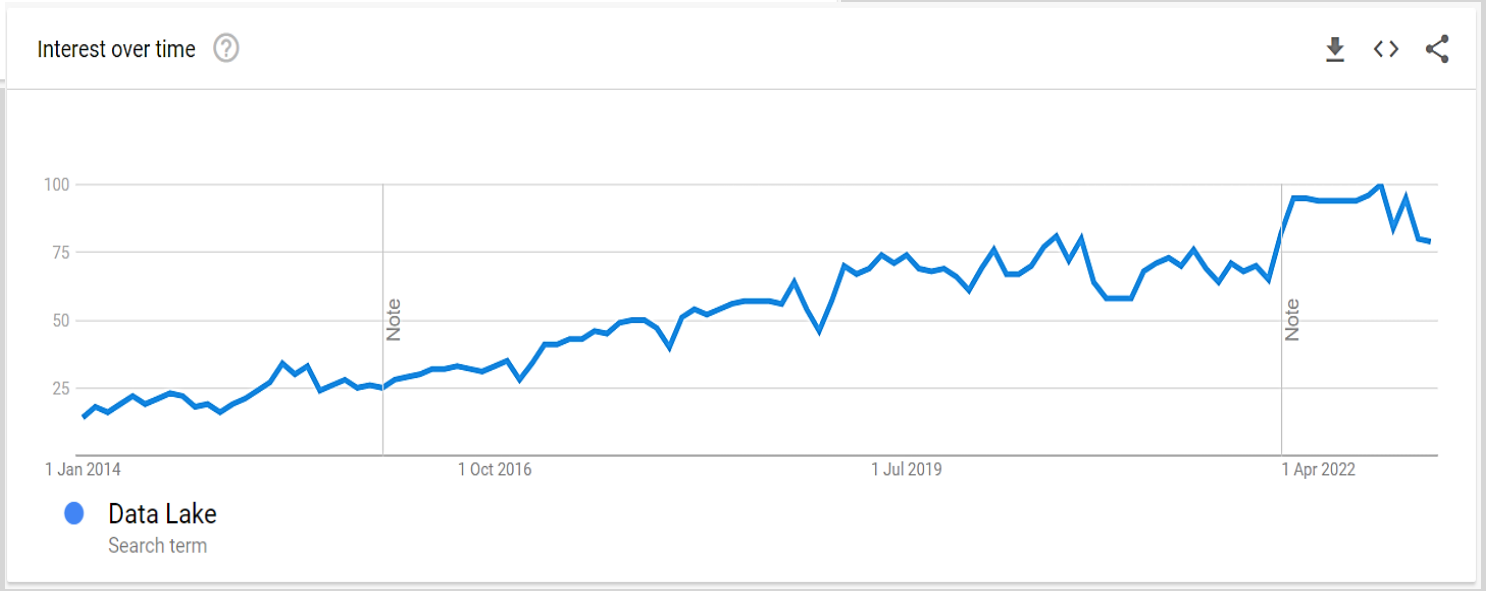
Data Storage Trend





Data Storage Trend





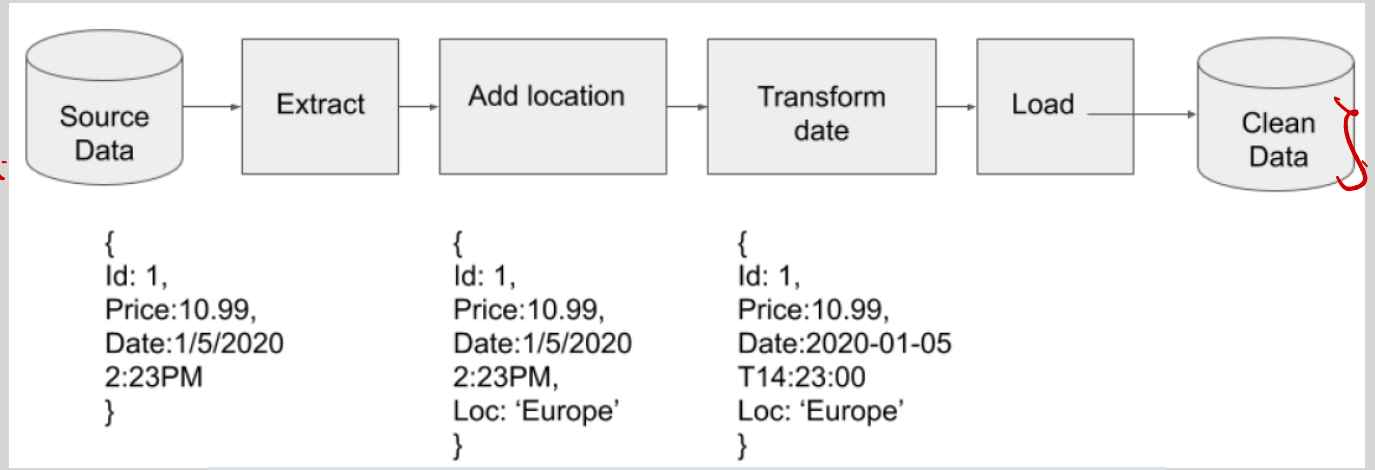
What Data Engineers Do





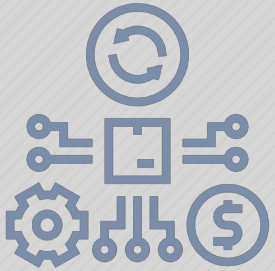
- Building data systems and pipelines.
  - The combination of extracting, loading, and transforming data is accomplished by creating a data pipeline.
  - Data pipeline → the design of systems for processing and storing data. These systems capture, cleanse, transform and route data to destination systems.

Data engineers build data pipelines that enable the organization to collect data points from millions of users and process the results in near real-time.



A pipeline that adds a location and modifies the date.

Data Pipeline Optimization



- Each of the ETL steps can be optimized:
  - Increasing the speed of data extraction.
  - Transformations can be optimized to improve join speeds to other tables and increase the performance of complex calculations or string manipulations.
  - Data loads can be optimized by increasing the speed at which data can be stored in the data store of choice or improving how the data is stored to be more effective for the end user.
- These optimizations are specific to the technology and final use case.
- The optimization is NOT a one-time task but an ongoing process that requires regular monitoring and tuning to maintain performance as the data and workloads change.

What Data Engineers Do





- Evaluating business needs and objectives
  - To make raw data useful to the organization, data engineers must understand business objectives.
  - Data engineers should understand business requirements and where data fits into the business model so they can build a data ecosystem that serves the organization's needs.



- Building algorithms and prototypes
  - Data pipelines represent an automated set of actions that extract data from various sources for analysis and visualization. These processes are powered by algorithms

Data Engineer Responsibilities

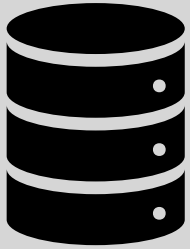


- Interpreting trends and patterns
  - Performing complex data analysis to find trends and patterns and reporting on the results in the form of dashboards, reports, and data visualizations
- Preparing data for prescriptive and predictive modeling
  - Data engineers must ensure the data is complete (no missing values), has been cleansed, and that rules have been established for outliers (eliminate, ignore, average out, and so on)
- Building algorithms and prototypes
  - Data pipelines represent an automated set of actions that extract data from various sources for analysis and visualization. These processes are powered by algorithms
- Developing analytical tools and programs

Data Engineering Skills



Coding. Proficiency in coding languages is essential to this role, so consider taking courses to learn and practice your skills. Common programming languages include SQL, NoSQL, Python, Java, R, and Scala.

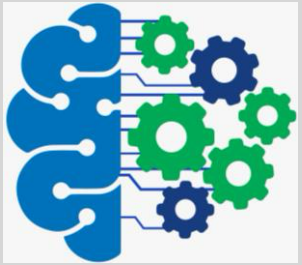


Data Storage. Selecting suitable data storage based on the business needs. Data engineers should be familiar with relational database and non-relational database, data warehouse, and data lake and how they work.

Data Engineering Skills



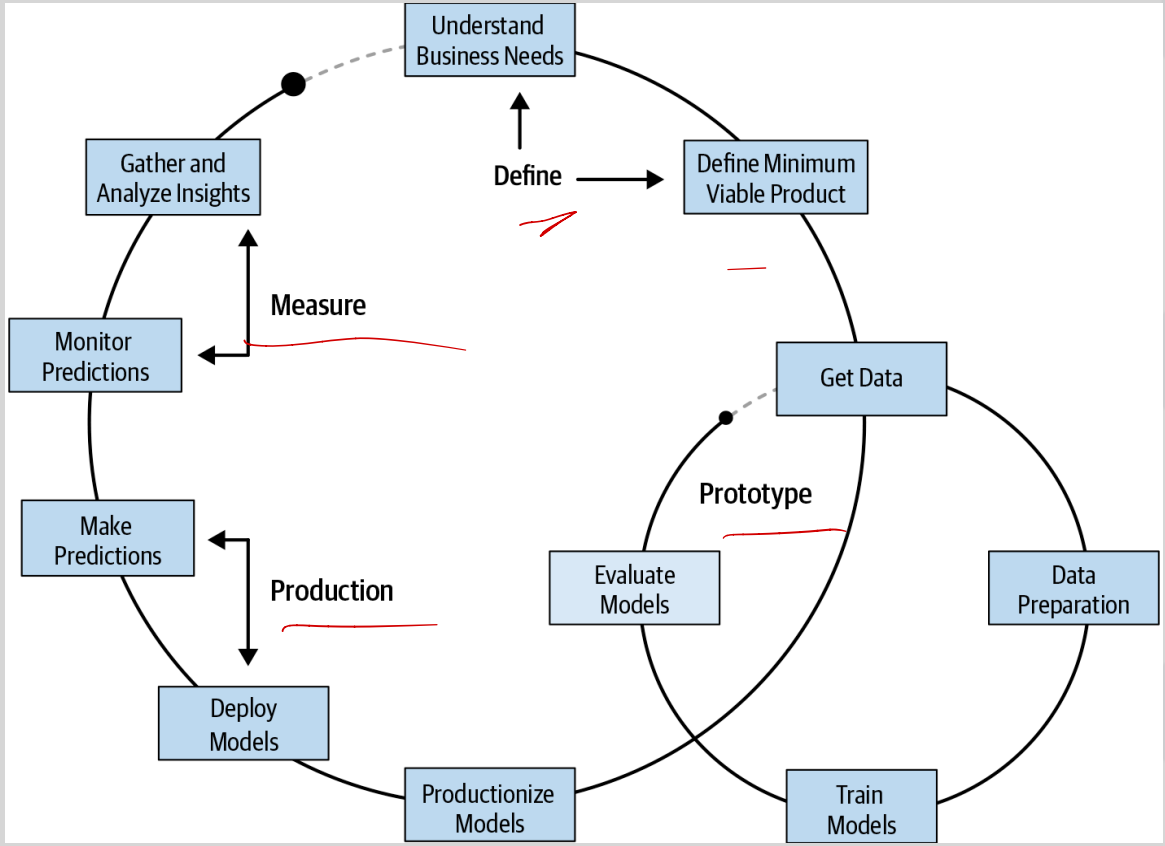
Data Pipeline. Data engineers should be able to build efficient data pipeline and system.



Machine learning. While machine learning is more the concern of data scientists, it can be helpful to grasp the basic concepts better to understand the needs of data scientists on the work team.

Data is the Backbone of AI and ML

- As the data scientists build a model, multiple iterations of data ingestion and preparation will be cycled through.
- As the data scientists ask more questions and require extra data for answers.
- The Data engineers and scientists could turn the code written to build the model into a little service that can be asked to run some data against the model or rebuild/enhance the model using new data.



Data Engineering Skills



Big data tools. Data engineers are tasked with managing big data. Tools and technologies vary by company, but some popular ones include Hadoop, MongoDB, and Kafka.



Cloud computing. Data engineers must understand cloud computing.



Data security. Data engineers and security teams are tasked with securely managing and storing data to protect it from loss or theft.

Data Engineering Tools



- Programming languages.
  - SQL, Java, Python, R.



- Databases.
  - Relational DB: MySQL, ORACLE, PostgreSQL.
  - NoSQL DB: MongoDB, Apache Cassandra, Elasticsearch.



- Data processing engines.
  - Data processing engines allow the parallel execution of transformation tasks.
  - Apache Spark.

Data Engineering Tools



- Data pipelines.
  - Combining a transactional database, a programming language, a processing engine, and a data warehouse results in a pipeline.
  - Data pipelines need a scheduler to allow them to run at specified intervals.
  - Apache Airflow & Apache NiFi → workflow management platforms.
    - Apache NiFi is designed to handle data in motion and provide data integration.
    - Apache Airflow is designed to handle data pipelines and workflow scheduling.

Both can be used together to achieve a complete data integration and management solution.

what is the difference between Apache Airflow and Apache NiFi?

Any Question

